From scalar-boundary Jacobi response to spectral positivity of regular de Sitter shells

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13 October 2026 — technical continuation draft

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Draft status. This is a separate technical continuation, not a submitted manuscript. The local even and completed odd kinetic calculations have received independent internal review. Global gauge, cone-flux and physical state-space assumptions remain explicit.

Abstract

We connect the scalar-boundary Jacobi response of a certified family of regular de Sitter shells to positivity of its radial scalar operators. A specific combination of cone Jacobi fields gives a nodeless zero-mass reference whose shell residual is strictly positive throughout the nonconstant branch $0 < \delta \leq 0.002$. For the reflection-even scalar problem, we construct the self-adjoint realization with a positive endpoint mass, prove the ground-state identity on its full closed form domain, and derive an explicit positive gap as a function of each fixed background. The reflection-odd matching instead gives an ordinary Robin realization with an analogous bound. The scalar essential threshold is $\mu^2 = 9/4$. A tensor partner comparison excludes every nonzero tensor spectral value below $9/4$ on the same backgrounds. These are applications of established spectral methods. Physical interpretation also requires the separately stated gauge, cone-domain, exceptional-harmonic and kinetic-action assumptions; no complete nonlinear stability or uniform positive physical mass gap is asserted.

1. Problem and relation to established methods

Stabilizing a static shell does not by itself exclude a scalar instability. The coupled Einstein–scalar equations on de Sitter slices contain a curvature shift, so positivity of the shifted radial eigenvalue is weaker than positivity of the four-dimensional mass [1,2]. Soft scalar junctions already lead to eigenparameter-dependent boundary conditions and boundary terms in the Rayleigh denominator [2–4]. Enlarged Hilbert spaces for such conditions [5], ground-state transformations [6], and factorized tensor operators [7,8] are established tools.

The contribution considered here is a particular exact application. The certified static branch supplies pointwise Jacobi signs. An explicit combination of those Jacobi fields solves the generic scalar perturbation equations at zero mass and has a strict shell residual. This additional structure, together with a closed-domain argument, upgrades a static response calculation to a statement

about the entire spectrum of specified radial operators. The decisive estimate does not divide an approximation error by the small central scalar slope and does not require an eigenvalue scan.

The accompanying bounded primary-source review compares these identities with prior scalar junction, radion, and tensor analyses. It does not establish exhaustive priority or global novelty of the mathematics. In particular, a positive radial Hilbert norm must be distinguished from a physical kinetic coefficient [9]. A doubled background also allows odd perturbations unless a reflection quotient is imposed; a calculation of the even sector alone does not settle that larger system [10,11].

The single-brane instability result of Lesgourgues and Sorbo [3] warrants a specific assumption comparison. Their setup has flat Minkowski slices, a compact orbifold and a smooth second fixed point with a nonzero warp factor and vanishing first scalar and warp derivatives. Their Eqs. (16)–(17) already contain the spectral parameter in the scalar junction. Here the other endpoint is a regular de Sitter cone with $\rho(0) = 0$, $a \sim 1/y$, and an infinite conformal throat. Their theorem therefore does not directly decide this boundary-value problem. The relevant distinction is the slicing and endpoint hypotheses, not the label "one brane"; the present application is not a counterexample to their theorem.

2. Exact background input and conventions

Use unit-curvature dS_4, with signature $(-+++)$, and two identical interiors joined at $y = T$:

$$ds^2 = dy^2 + \rho(y)^2 \gamma_{\mu\nu} dx^\mu dx^\nu, \qquad p = \eta', \quad a = \rho'/\rho, \quad H = \rho^{-2}. \tag{1}$$

The action includes the boundary completion,

$$S = \sum_{i=\pm} \left[\int_{M_i} \sqrt{-g} \left(\frac{R}{2} - \frac{1}{2}(\nabla\eta)^2 - U \right) + \int_\Sigma \sqrt{-h}\, K_i \right] - \int_\Sigma \sqrt{-h}\, \sigma(\eta). \tag{2}$$

Both shell normals point toward increasing radius in their own interiors. The background equations and identical-copy junctions are

$$\begin{aligned} p' &= U' - 4ap, \qquad a' = -H - p^2/3, \quad a^2 = H + p^2/12 - U/6, \\ a(T) &= \sigma/6, \qquad\quad p(T) = -\sigma'/2. \end{aligned} \tag{3}$$

Primes on potentials denote field derivatives.

The registered model is

$$\begin{aligned} W(\eta) &= \frac{1}{3} + \eta^2 + \frac{\eta^3}{3}, \qquad\qquad U = \frac{1}{2}(W')^2 - \frac{2}{3}W^2, \\ \sigma &= 2W + \delta(1 + c + c\eta), \quad c = \frac{2}{1.0357712571566784} - \frac{4}{3} = \frac{96714321425692}{161839258930731}. \end{aligned} \tag{4}$$

The decimal in (4) is interpreted exactly as a rational number. The October 11 static theorem supplies, for every $0 < \delta \le 1/500$, one regular solution in its specified small Dirichlet tube, with $T \ge 9$, $|\eta| \le 1/100$, $a > 0$, and $p < 0$ on $(0, T]$. Uniqueness is restricted to that tube. The theorem is a mathematical branch statement; it does not certify historical floating-point trajectories individually.

The field shift $\eta = \phi - 1$ rewrites the earlier $W_{\rm old}(\phi) = 1 - \phi + \phi^3/3$ exactly as (4). The previously investigated unstable hilltop configuration and the regular branch treated here are distinct

backgrounds of the same action: the former has a cone near $\phi = -1$, whereas this certified tube is near $\phi = +1$. Tube-local uniqueness does not exclude the former background. No dynamical evolution connecting the branches is established. By contrast, replacing the affine detuning by a scalar-independent detuning changes the boundary theory; that earlier curvature comparator is not another branch of the same shell action.

Write $H_b^2 = \rho(T)^{-2}$. Our harmonic convention is

$$\Box_\gamma Y = \mu^2 Y, \qquad m_4^2 = H_b^2 \mu^2, \qquad \Lambda = \mu^2 + 4. \tag{5}$$

Thus exclusion of nonpositive scalar masses requires exclusion of $\Lambda \le 4$, not merely $\Lambda \le 0$. The constant-scalar $\delta = 0$ limit is outside the divided scalar formulation below.

3. The Jacobi reference and its strict shell residual

Let $u = \eta(0)$ label the regular cone family at fixed radial coordinate, and define

$$h = \eta_u, \quad q = p_u = h', \quad v = (\log \rho)_u, \quad b = a_u = v', \quad j = b + ph/3, \quad D = hp' - pq. \tag{6}$$

The static proof establishes $h \ge 1$, $q > 0$, $pj > 0$, and $pD > 0$ on a nonconstant interior. It also gives

$$J = \frac{q(T)}{h(T)} + \frac{\sigma''}{2} \ge 1, \qquad B_\partial = \frac{p'(T)}{p(T)} + \frac{\sigma''}{2} > J, \quad B_\partial - J = \frac{D(T)}{p(T)h(T)}. \tag{7}$$

The relevant constrained Jacobi identities are

$$j' + 4aj = -2Hv, \qquad D' + 4aD = 4p(Hh + pj), \qquad pq - p'h = 12aj + 12Hv. \tag{8}$$

For a generic scalar harmonic, the longitudinal gauge-invariant fields (Ψ, X) obey

$$\Psi' = -2a\Psi - pX/3, \qquad pX' = p'X + (3\Lambda H - 2p^2)\Psi. \tag{9}$$

Set

$$K = \rho^2 j, \qquad X_0 = h + pK, \qquad \Psi_0 = v + aK = -\rho^2 D/12. \tag{10}$$

Equation (8) gives $K' = -2\Psi_0$; substitution proves both equations (9) at $\Lambda = 4$. Thus (10) is an actual zero-mass bulk solution, not the uncorrected static variation (h, v). Moreover $X_0 > h \ge 1$ and $p\Psi_0 < 0$. On the registered branch $\Psi_0 > 0$, so $P_0 = \rho^2 \Psi_0$ is positive throughout the interior.

The full even scalar shell mismatch is

$$M = X' + \frac{\sigma''}{2} X + 2p\Psi. \tag{11}$$

Retaining its gravitational term, (10) gives

$$M_0 = hJ + B_\partial pK, \qquad \frac{M_0}{X_0} = \frac{J + B_\partial \alpha_K}{1 + \alpha_K} > J \ge 1, \qquad \alpha_K = \frac{pK}{h} > 0. \tag{12}$$

This is a normalization-independent strict margin. It requires no lower bound on the small central slope. The cone expansions are

$$p = \frac{U'(u)}{5} y + O(y^3), \quad X_0 = 1 + O(y^2), \quad \Psi_0 = -\frac{U'(u)}{60} y^2 + O(y^4), \quad P_0 = -\frac{U'(u)}{60} y^4 + O(y^6). \tag{13}$$

The static junction determinant is also positive: $\det D_{(u,T)}(a - \sigma/6, p + \sigma'/2) = H_b M_0 > 0$. Nodelessness follows from the pointwise signs in (10), separately from that determinant.

4. A closed scalar realization and a positive gap

4.1 Endpoint mass and form domain

For $P = \rho^2\Psi$, define

$$\alpha = \frac{1}{\rho^2 p^2}, \qquad q_{SL} = \frac{2}{3\rho^2}, \qquad w_{SL} = \frac{1}{\rho^4 p^2}, \qquad \beta_{SL} = \rho(T)^4 p(T)^2 B_\partial > 0. \tag{14}$$

Here α is the Sturm–Liouville leading coefficient, not the background momentum. Equations (9) and (11) give

$$-(\alpha P')' + q_{SL} P = \Lambda w_{SL} P, \qquad \alpha(T)P'(T) = \frac{\Lambda}{\beta_{SL}} P(T). \tag{15}$$

In particular $P' = -\rho^2 p X/3$. The reference residual is

$$r_0 = \frac{\alpha(T)P_0'(T)}{P_0(T)} - \frac{4}{\beta_{SL}} = -\frac{M_0}{3\rho(T)^2 p(T) B_\partial \Psi_0(T)} > 0. \tag{16}$$

The even Hilbert space is

$$\mathscr{H}_e = L^2(w_{SL}dy) \oplus \mathbb{C}, \qquad \|(P,z)\|_e^2 = \int_0^T w_{SL}|P|^2 dy + \frac{|z|^2}{\beta_{SL}}. \tag{17}$$

Its components are independent. Trace identification $z = P(T)$ belongs to the form domain

$$V_e = \left\{ (P, P(T)) : P \in AC_{\mathrm{loc}}((0,T]),\ \int_0^T (w_{SL}|P|^2 + \alpha|P'|^2)dy < \infty \right\}. \tag{18}$$

The energy is $\mathfrak{e}[P] = \int(\alpha|P'|^2 + q_{SL}|P|^2)$. Since q_{SL}/w_{SL} is bounded, it introduces no additional integrability condition. Local H^1 convergence proves closedness and identifies the shell trace. Narrow shell bumps establish density in the enlarged Hilbert space.

Smooth functions vanishing near the cone, with unrestricted shell trace, form a core. For a cutoff over $[\epsilon, 2\epsilon]$, its extra derivative energy is bounded by a constant times the weighted L^2 tail, because $\alpha/w_{SL} = \rho^2 = O(y^2)$. The representation theorem therefore gives the self-adjoint Friedrichs operator

$$A_e(P, P(T)) = \left(w_{SL}^{-1}[-(\alpha P')' + q_{SL}P],\ \beta_{SL}\alpha(T)P'(T)\right). \tag{19}$$

Its domain is (18) with $\alpha P'$ locally absolutely continuous and the first component in $L^2(w_{SL}dy)$. Equation (15) is precisely its eigenvalue equation.

At $\Lambda = 4$, reduction of order gives a second solution $P_0 \int(\alpha P_0^2)^{-1}dy \sim y$; $P_0 \sim y^4$. Only the latter is in $L^2(w_{SL}dy)$, since $w_{SL} \sim A_c^{-2} y^{-6}$, $A_c = U'(u)/5$. The cone is limit-point. No independent cone boundary parameter is introduced.

4.2 Ground-state identity and quantitative coercivity

Let

$$a_g = \alpha P_0^2, \quad m_g = w_{SL} P_0^2, \quad b_g = P_0(T)^2/\beta_{SL}, \quad k_g = r_0 P_0(T)^2 > 0. \tag{20}$$

For $P = P_0 f$, the closed ground-state identity is

$$\mathfrak{e}[P] - 4\|(P, P(T))\|_e^2 = \int_0^T a_g |f'|^2 dy + k_g |f(T)|^2 =: D[f]. \tag{21}$$

It holds first on the cone core by integration by parts. On that core, the original form norm equals $D + 5\|P\|_e^2$; the same cutoff argument, now using $a_g/m_g = \rho^2$, proves the identity on the full form domain. This avoids assigning Frobenius expansions to arbitrary form vectors.

Strict positivity of (21) for each nonzero vector would not alone exclude zero from continuous spectrum. A weighted Hardy estimate supplies the stronger conclusion. Define

$$\alpha_* = \inf_{0<y\le T} \frac{a_g(y)}{y^4} > 0, \quad m_* = \sup_{0<y\le T} \frac{m_g(y)}{y^2} < \infty, \quad M_g = \int_0^T m_g dy. \tag{22}$$

The limits $a_g/y^4, m_g/y^2 \to 1/144$, together with positivity on compact subintervals, establish these bounds at each fixed background. For $g = f - f(T)$, integration by parts on $[\epsilon, T]$ gives

$$3\int_\epsilon^T y^2|g|^2 = -\epsilon^3|g(\epsilon)|^2 - 2\Re\int_\epsilon^T y^3\bar{g}g' \le 2\left(\int_\epsilon^T y^2|g|^2 \int_\epsilon^T y^4|g'|^2\right)^{1/2}. \tag{23}$$

Hence $\int m_g|f - f(T)|^2 \le 4m_*/(9\alpha_*)\int a_g|f'|^2$. The Hilbert-space triangle and Cauchy–Schwarz inequalities then yield

$$\boxed{A_e - 4I \ge \varepsilon_e I, \qquad \varepsilon_e = \left[\frac{4m_*}{9\alpha_*} + \frac{M_g + b_g}{k_g}\right]^{-1} > 0.} \tag{24}$$

Thus the entire shifted spectrum, including its continuous part, lies above a positive number. Self-adjointness also excludes complex spectral values. Formula (24) is an explicit function of a background; no numerical value or lower bound uniform in δ has been evaluated here. Its value is unchanged by rescaling P_0.

4.3 Essential spectrum and odd Robin realization

Assume the regular-cone expansions through two logarithmic derivatives. The Liouville map

$$x = \int_y^T \frac{dt}{\rho(t)}, \qquad F = \frac{P}{\rho^{3/2}|p|}, \qquad s_L = \rho\left(\frac{3}{2}a + \frac{p'}{p}\right) \tag{25}$$

puts the cone at $x = \infty$ and the shell at $x = 0$. The transformed potential is

$$V_L = \frac{2}{3}\rho^2 p^2 + (s_L)_x + s_L^2 = \frac{25}{4} + O(e^{-2x}), \quad -F_x(0) + s_L(0)F(0) = \frac{\Lambda}{\rho(T)B_\partial}F(0). \tag{26}$$

The endpoint mass remains positive. A decaying bounded potential perturbation is relatively compact on the half-line. Comparing the endpoint realization with a Dirichlet operator having the same actual potential changes its resolvent by finite rank. Consequently

$$\sigma_{\mathrm{ess}}(A_e - 4I) = [9/4, \infty). \tag{27}$$

Any spectrum below that threshold is discrete. The best actual gap cannot exceed $9/4$.

If both interiors fluctuate independently, induced scalar and metric continuity give $X_o + p\xi_o = 0$ and $\Psi_o + a\xi_o = 0$. The generic summed trace-free junction fixes $\xi_e = 0$. Thus the even mismatch is (11), whereas odd continuity implies

$$\alpha(T)P_o'(T) + c_o P_o(T) = 0, \qquad c_o = \frac{1}{3a(T)\rho(T)^2} > 0. \tag{28}$$

The odd Hilbert norm has no endpoint mass: $\|P\|_o^2 = \int w_{SL}|P|^2$. Its form includes $+c_o|P(T)|^2$. With $k_o = [\alpha P_0'/P_0 + c_o]_T P_0(T)^2 > 0$, the same proof gives

$$A_o - 4I \geq \varepsilon_o I, \qquad \varepsilon_o = \left[\frac{4m_*}{9\alpha_*} + \frac{M_g}{k_o}\right]^{-1} > 0, \quad \sigma_{\rm ess}(A_o - 4I) = [9/4, \infty). \tag{29}$$

This operator statement assumes the displayed two-copy matching and its finite-energy realization. It does not identify the odd radial norm with a physical kinetic norm by definition.

5. Exceptional harmonic subspaces and physical kinetic information

5.1 Covariant exceptional classification

The generic reduction uses both $D_\mu Y$ and the trace-free Hessian $T_{\mu\nu}[Y] = D_\mu D_\nu Y - \gamma_{\mu\nu}\Box_\gamma Y/4$. Their degeneracies require separate analysis [12]. On unit dS_4,

$$D^\mu T_{\mu\nu}[Y] = \frac{3}{4}(\mu^2 + 4) D_\nu Y. \tag{30}$$

Thus $DY = 0$ identifies the constant subspace at $\mu^2 = 0$. If $T[Y] = 0$ but $DY \neq 0$, then $\mu^2 = -4$ and $D_\mu D_\nu Y = -\gamma_{\mu\nu} Y$. Other harmonics at these same mass values remain generic. For example, the homogeneous de Sitter exponents $r = 0, 1$ are exceptional, while $r = -3, -4$ have the same respective masses and are generic.

For the constant sector, assume the regular static Jacobi family spans the admitted perturbations modulo radial gauge. The even junction determinant is $H_b M_0 > 0$. The odd induced-field matrix has determinant

$$\det \begin{pmatrix} h & p \\ v & a \end{pmatrix} = ah - pv = aX_0 - p\Psi_0 > 0. \tag{31}$$

There is therefore no constant modulus within that complete regular family. For the Hessian-degenerate sector, the local bulk spanning calculation is pure gauge. If the required transformations are globally admissible, gauge the bulk fields to zero and retain the two embedding perturbations $\Delta_\pm$. Scalar continuity and the summed scalar junction give

$$p(\Delta_+ - \Delta_-) = 0, \qquad pB_\partial(\Delta_+ + \Delta_-) = 0. \tag{32}$$

Their determinant has magnitude $2p^2 B_\partial > 0$, excluding the relative embeddings. These conclusions depend on the stated completeness and global gauge assumptions; they are not obtained by substituting exceptional mass values into a divided generic equation.

5.2 Local even and completed odd kinetic coefficients

The quadratic action, including Einstein–Hilbert, GHY and shell terms, must decide the kinetic sign. In a zero-shift radial ADM decomposition, the tangential two-derivative density before the generic constraint is $\rho^2[3DN\cdot D\psi + 3(D\psi)^2 - (DX)^2/2]$. On the longitudinal constraint surface $N = -2\Psi$, the doubled even kinetic coefficient is

$$K_e = 2\int_0^T \rho^2(|X|^2 + 6|\Psi|^2)dy = 18\mathfrak{e}[P] = 18\Lambda\|(P, P(T))\|_e^2. \tag{33}$$

The last equality holds at spectral roots. The first equality is a local action calculation with a straight shell; the middle equality follows from $P' = -\rho^2 pX/3$. The factor Λ is essential: the

radial norm alone is not the physical kinetic coefficient. This gives the healthy kinetic sign for finite-energy nonzero generic even profiles with the admitted longitudinal representative.

An odd profile has a moving shell in longitudinal gauge. The bulk calculation alone gives

$$K_{o,\mathrm{bulk}} = 18\Lambda \int w_{SL}|P_o|^2 dy - 18c_o|P_o(T)|^2. \tag{34}$$

The completed graph-boundary calculation includes the terms missing from that bulk expression. Write the shell as $y = T + \epsilon\zeta Q(x)$, with odd continuity $\psi_b = -a_b\zeta$, $X_b = -p_b\zeta$. The tangential Einstein divergence on a moving radial domain supplies a shell flux. Together with graph GHY minus coordinate-slice GHY, its one-copy quadratic action coefficient is $+3a_b\rho_b^2\zeta^2(DQ)^2$. The shifted upper limit of the radial ADM intrinsic-curvature term supplies $+3\rho_b^2\zeta\psi_b(DQ)^2$ per copy. The one common shell volume contributes $-\sigma_b\rho_b^2\zeta^2(DQ)^2/2$.

Adding both gravitational copies and one shell, then using $\sigma_b = 6a_b$ and continuity, gives the boundary action coefficient $-3a_b\rho_b^2\zeta^2(DQ)^2$. Thus

$$K_{o,\partial} = 6a_b\rho_b^2\zeta^2 = 18c_o|P_o(T)|^2, \qquad \boxed{K_o = 18[\mathfrak{e}[P_o] + c_o|P_o(T)|^2] = 18\Lambda \int_0^T w_{SL}|P_o|^2 dy.} \tag{35}$$

The last equality again holds at spectral roots. The two gravitational edge terms cancel after odd continuity is imposed; this explains why the final contribution equals the shell-volume value. Assuming that equality before calculating the moving Einstein and GHY terms would miss essential sign checks.

Compactly supported tangential test variations justify the integrations by parts. They vanish near auxiliary initial and final time surfaces, where the graph therefore has no perturbed corner angle. This determines the local kinetic bilinear without assuming a moving graph is orthogonal to arbitrary time boundaries. An independent covariant graph derivation agrees with the direct normal calculation.

Equations (33) and (35) supply healthy local kinetic signs for generic admitted finite-energy profiles. They do not prove that every physical state has a regular longitudinal representative or that the global covariant symplectic flux has the prescribed cone limit. Those state-space and gauge questions remain distinct from the closed radial form theorem.

6. Tensor partner bound throughout the same branch

The pure transverse-traceless reduction is $h'' + 4ah' + \mu_T^2\rho^{-2}h = 0$. Two-copy tensor continuity and the sum junction give even Neumann and odd Dirichlet conditions. With

$$dz = dy/\rho, \quad z(T) = 0, \quad \psi = \rho^{3/2}h, \quad S = \frac{3}{2}\rho', \quad D = \partial_z - S, \quad D^* = -\partial_z - S, \tag{36}$$

the original tensor operator is D^*D. The tensor norm is $\int \rho^2|h|^2dy = \int |\psi|^2dz$. Factorization and this massless profile are standard [7]; a de Sitter continuum threshold $3H_b/2$ is also established in the scalar-free setting [8].

The useful comparison on the nonconstant branch is

$$V_- = S^2 - S_z = \frac{9}{4} + \rho^2\left(\frac{9}{16}p^2 - \frac{1}{8}U\right) \geq \frac{9}{4} \qquad (U \leq 0). \tag{37}$$

For even fields, take $\operatorname{Dom} D = H^1(-\infty, 0]$; the adjoint has zero shell trace. The original condition is $D\psi(0) = 0$, and the partner DD^* is Dirichlet. For odd fields, reverse these first-order domains. The partner condition is $\chi'(0) + S_b\chi(0) = 0$, whose form has the positive endpoint term $S_b|\chi(0)|^2$, since $S_b > 0$. Both partner forms are bounded below by $9/4$.

The polar decomposition of the closed first-order operator compares the entire nonzero spectra of D^*D and DD^*, with their correct domains [13]. The even kernel is the normalizable $\rho^{3/2}$; the odd Dirichlet condition removes it. Therefore the even tensor spectrum is $\{0\} \cup [9/4, \infty)$, and the odd spectrum is $[9/4, \infty)$ as spectral sets. Both have essential spectrum $[9/4, \infty)$; this statement does not need an additional claim about embedded eigenvalues or threshold spectral type.

The model gate in (37) is exact. Expanding (4),

$$U = -\frac{2}{27} + \frac{14}{9}\eta^2 + \frac{50}{27}\eta^3 - \frac{1}{6}\eta^4 - \frac{4}{9}\eta^5 - \frac{2}{27}\eta^6. \tag{38}$$

On $|\eta| \le 1/100$, its coefficient upper bound is

$$U \le U(0) + \sum_{j\ge1} |U_j|(1/100)^j = -\frac{997874976899}{13500000000000} < 0. \tag{39}$$

The comparison thus applies to every background in the certified interval. Nonzero physical tensor masses satisfy $m_T \ge 3H_b/2$. This is a reduced tensor conclusion, not a complete ghost analysis of all gravitational and shell degrees of freedom.

7. Evidence, limitations and reproducibility

The mathematical conclusions are full-spectrum statements for the specified finite-energy radial realizations. The physical conclusions additionally require the matching conditions, mode reductions, and admissible global gauge/cone class described above. In particular, neither normalizability in the chosen radial norm nor a local pure-gauge calculation disposes of externally imposed cone data or every global extension beyond the cone.

The scalar gap formulas (24) and (29) are positive for each fixed $\delta > 0$. They have not been numerically evaluated or bounded uniformly in dimensionless units. In physical units the essential threshold is $(9/4)H_b^2$, and $H_b^2 \to 0$ on the branch as $\delta \to 0$. A strictly positive physical mass floor uniform over that limit is therefore incompatible with this spectral family. The tensor dimensionless threshold is exactly controlled, while its physical scale also follows H_b.

The supplementary domain proof gives the full trace, closure, limit-point, Hardy and essential-spectrum arguments. Its portable algebra checker retains ten original identities and adds full spectral-shift and boundary checks, for thirteen positive checks and four deliberately wrong formulas with exact nonzero witnesses. Normal and optimized Python produce byte-identical receipts; injected failures remain failures in both modes. The tensor checker passes eight identities/sign checks and rejects four wrong formulas. The completed odd action has eleven exact checks, including two wrong-formula controls, and a separate covariant reconstruction. These computations support algebra and exact polynomial arithmetic. They do not constitute formal verification of functional analysis or physical completeness. Independent internal reviews inspect those arguments separately; they are not external peer review.

No broad floating-point spectral scan is used to prove the new inequalities. No claim is made about vector-sector completeness, nonlinear attraction, vacuum decay, reheating, or cosmological observables.

Source and data availability

The October 11 static theorem and exact arithmetic certificate are the dated background basis. The October 11 Jacobi-to-zero-mode bridge is identified separately as a dated prospective calculation. The October 13 domain proof, completed operator audit, tensor comparison, literature review, exact checkers and receipts form the new dated supplementary record.

The prior static basis is available at the immutable October 11 source commit. The capsule contains an unchanged static basis, the dated prospective bridge, and source identities. The continuation's relative manifest identifies its contents; external release metadata binds the final source commit.

The domain manifest pins the unchanged proof and distinguishes original-authoring hashes from adapted CLI hashes. Historical public and private source files, including the previously approved manuscript packet, are unchanged.

Declarations

Funding: No funding was received for this work.
Competing interests: The author declares no known competing interests.
AI assistance: Substantial OpenAI Codex assistance was used for mathematical exploration and drafting, symbolic-check implementation, literature organization, and internal reviews by separate agent processes. These reviews do not substitute for human scientific judgment or external peer review. The author retains responsibility for the scientific content; this draft remains subject to author review.

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